

Ab initio investigation of α - and ζ -V₂O₅ for beyond lithium ion battery cathodes

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Section S1:. α -V₂O₅ Empty Properties

S1.1. Physical

Several computational studies [1–9] have reported lattice parameters for empty α -V₂O₅ and are included in Table S1. However, only the PBE data (bold) matches the methods of this investigation and direct comparison could only be made to those data. When comparing to these data, our results yield a mean absolute error (MAE) of 0.01 Å, 0.10 Å, and 0.02 Å for the a , b , and c lattice parameters, respectively. Overall, these parameters range from 3.57–3.63 Å, 4.54–4.86 Å, and 11.54–11.60 Å when PBE was utilized. Other functional levels, both of lower and higher levels of theory, are included merely to show the variation between computational studies (i.e., direct comparison should not be made to them). For all functional levels, the lattice parameters range from 3.36–3.67 Å, 4.19–4.86 Å, and 11.48–11.86 Å, respectively. Although across-the-board comparison should not be made, our structures are in excellent agreement with other computational studies.

Functional Level	Lattice Parameter			Ref.
	a(Å)	b(Å)	c(Å)	
LDA	3.564	4.373	11.519	[1]
PBE	3.567	4.703	11.550	^a
PBE	3.57	4.83	11.54	[2]
PBE	3.57	4.83	11.54	[3]
PBE	3.565	4.696	11.563	[4]
PBE	3.632	4.682	11.571	[5]
PBE	3.57	4.54	11.60	[6]
PBE	3.567	4.858	11.542	[7]
PBE+ U	3.637	4.798	11.483	[4]
PBE+ U	3.360	4.804	11.496	[7]
PBE+ U	3.636	4.790	11.489	[8]
vdW-D2	3.585	4.478	11.860	[4]
vdW-D2	3.665	4.593	11.746	[8]
PBE-D2	3.53	4.47	11.64	[2]
PBE-D2	3.54	4.40	11.64	[6]
PBE-D3	3.553	4.378	11.638	^a
TS	3.54	4.35	11.67	[3]
TS+SCS	3.56	4.51	11.59	[3]
PW91	3.600	4.401	11.532	[9]
PBEsol	3.54	4.32	11.55	[6]
optPBE-vdW	3.628	4.423	11.690	[5]
optB88-vdW	3.622	4.224	11.741	[5]
optB86b-vdW	3.621	4.188	11.715	[5]

Table S1: Lattice parameters for empty α -V₂O₅ from this work^a and other computational works. $U = 4.0$ eV for all PBE+ U entries.

S1.2. Electronic

Partial or atom-resolved DOS were completed for the empty α -V₂O₅ structure using the xc functionals and vdW corrections listed in Sec. 1, with the corresponding bandgaps for each plot shown in Fig. 1b. When comparing LDA, PBE, and the PBE+ U plots, an expected steady increase in bandgap can be seen. What is unexpected is how the meta-GGA and hybrid functionals (SCAN and HSE06, respectively) result in bandgaps in agreement with standard GGA functionals. Based on the behavior of these higher level of theory functionals, later electronic calculations (i.e., those for intercalated systems) were investigated with only PBE and PBE-D3.

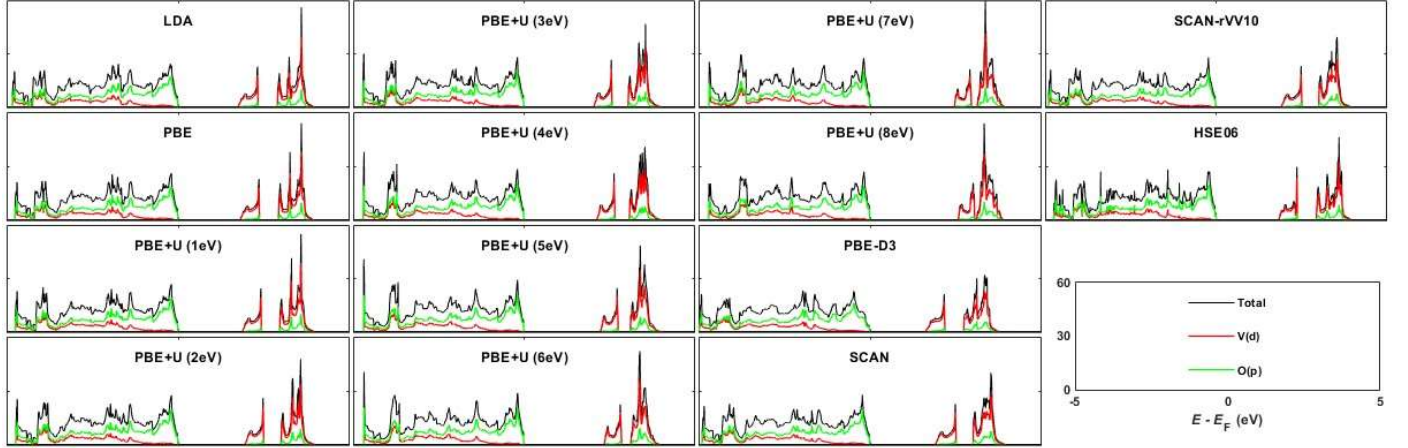


Figure S1: Atom-resolved DOS for empty α -V₂O₅ using various functionals. The total DOS are plotted in black, vanadium d -states in red, and oxygen p -states in green. The x -axis is zoomed to $[-5,5]$ and adjusted so that “0” corresponds to the Fermi energy.

Functional	Bandgap (eV)	Ref.
Exp.	2.2	[10]
Exp.	2.35	[11]
Exp.	2.30	[12]
Exp.	2.24	[13]
Exp.	2.38	[14]
Exp.	2.42	[15]
Exp.	2.49	[15]
Exp.	2.35	[16]
Exp.	1.7	[17]
Exp.	2.1	[17]
LDA	1.735	^a
LDA	1.74	[18]
LDA	1.9	[19]
LDA	2.3	[19]
LDA	2.5	[20]
PBE	1.795	^a
PBE	1.84	[17]
PBE+ U ($U = 1.0$ eV)	1.860	^a
PBE+ U ($U = 2.0$ eV)	1.940	^a
PBE+ U ($U = 3.0$ eV)	2.025	^a
PBE+ U ($U = 3.1$ eV)	2.3	[21]
PBE+ U ($U = 4.0$ eV)	2.125	^a
PBE+ U ($U = 5.0$ eV)	2.223	^a
PBE+ U ($U = 6.0$ eV)	2.340	^a
PBE+ U ($U = 7.0$ eV)	2.460	^a
PBE+ U ($U = 8.0$ eV)	2.555	^a
PBE-D3	1.595	^a
SCAN	1.895	^a
SCAN-rVV10	1.895	^a
HSE06	1.805	^a

Table S2: Reported bandgaps for empty α -V₂O₅ from this work^a, as well as experimental and other computational works.

Section S2:. α -V₂O₅ Intercalated Properties

S2.1. Physical

Lattice parameters for the mono-, di-, and trivalent ion intercalation described in Sec. 2.2 are listed in Table S3. Li_{0.17} and Al_{0.20} have similar lattice parameters for PBE, differing by a few hundredths of an Å and show a more discernable difference in b and c with PBE-D3 (0.16 Å for both). The subsequent concentration significantly differs between Mg_{0.33} and Ca_{0.33} for lattice parameter b (0.13 Å) with PBE. However, these two intercalations are similar with PBE-D3 and Li_{0.33} differs in b by 0.11 Å and 0.14 Å from each, respectively. Finally, Li_{0.67} and Na_{0.67} possess the largest difference found. These two differ in b by 0.82 Å when using PBE-D3, but differ in b and c in PBE (2.72 Å and 0.74 Å, respectively). Due to the nature of the Li ion (i.e., lightest ion investigated), it is believed the forces acting on it are able to move the ion between polyhedra enclosed by the layers, which in turn cause the layer separation.

No clear trend in lattice parameters is evident through Li intercalation concentrations. However, some similarities emerge when comparing each concentration to the empty structure rather than the preceding concentration. For both PBE and PBE-D3, b is always larger upon intercalation while c is always smaller. Lattice parameter a is consistently larger for PBE-D3, while it is smaller in Li_{0.66} for PBE. When using PBE the VO₅ polyhedra distort as Li concentration increases, causing a “puckering” to occur in the layers. This behavior is indicative of the $\alpha \rightarrow \epsilon$ structural change seen in lithiated V₂O₅ [22]. These distortions are less discernable when using PBE-D3, however.

Cell volume is also included in Table S3. PBE-D3 follows an expected pattern of higher concentration yields larger volume and larger ions intercalated at the same concentration result in larger volume. PBE, however, has anomalies. Li_{0.33} is smaller than the preceding Li concentration and Mg_{0.33} is smaller than its corresponding Li intercalation. Al_{0.20} is difficult to analyze as it is smaller than Li, but at a higher concentration. Therefore, the lower volume may or may not be such an anomaly.

M _x	Lattice Parameter			Vol.(Å ³)
	a(Å)	b(Å)	c(Å)	
V ₄ O ₁₀ (PBE)	3.567	4.703	11.550	193.729
Li _{0.17}	3.574	4.999	11.442	204.459
Li _{0.33}	3.569	4.892	11.413	199.274
Li _{0.50}	3.595	5.274	11.338	214.896
Li _{0.67}	3.535	5.503	11.536	224.383
Na _{0.67}	3.613	8.219	10.795	320.539
Mg _{0.33}	3.561	4.849	11.368	196.293
Ca _{0.33}	3.591	4.978	11.376	203.392
Al _{0.20}	3.538	4.907	11.421	198.288
V ₄ O ₁₀ (PBE-D3)	3.553	4.378	11.638	181.057
Li _{0.17}	3.562	4.501	11.539	184.989
Li _{0.33}	3.567	4.560	11.411	185.591
Li _{0.50}	3.573	4.890	11.345	198.217
Li _{0.67}	3.562	5.049	11.455	205.973
Na _{0.67}	3.565	5.870	11.394	238.458
Mg _{0.33}	3.561	4.670	11.330	188.389
Ca _{0.33}	3.593	4.695	11.367	191.769
Al _{0.20}	3.554	4.660	11.380	188.506

Table S3: Lattice parameters for the various M_x concentrations in α -V₂O₅ when using PBE and PBE-D3. All structures remained orthorhombic. NOTE: concentrations are relative to V₂O₅, yet lattice parameters are relative to the V₄O₁₀ unit cell as shown in Fig. 1a.

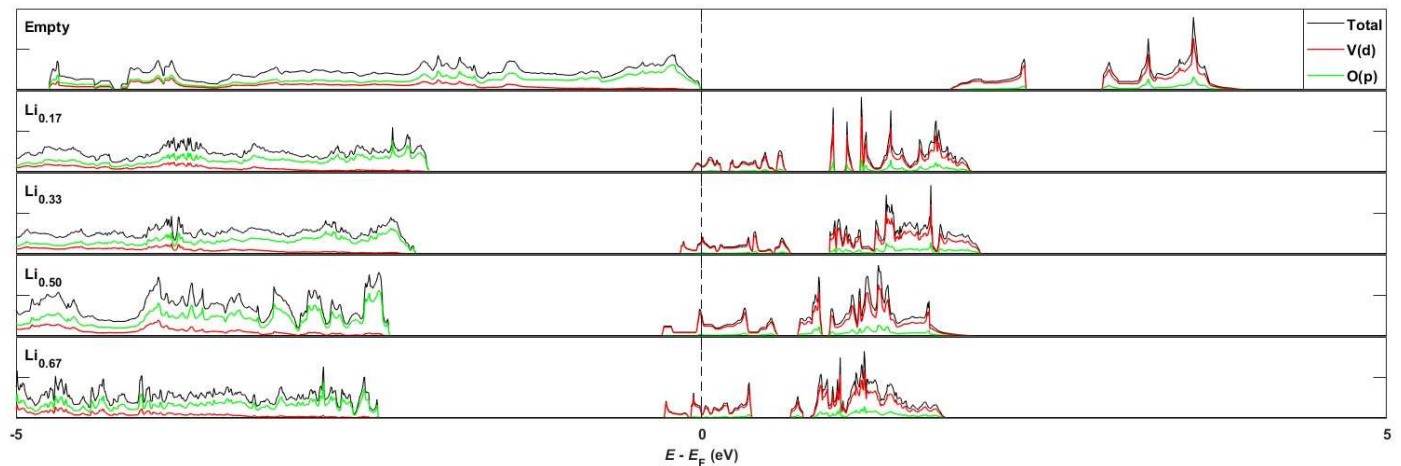


Figure S2: Atom-resolved DOS for Li intercalated in α -V₂O₅ using PBE. The total DOS are plotted in black, vanadium *d*-states in red, and oxygen *p*-states in green. The *x*-axis is zoomed to $[-5, 5]$ and adjusted so that “0” corresponds to the Fermi energy, denoted by the dashed line.

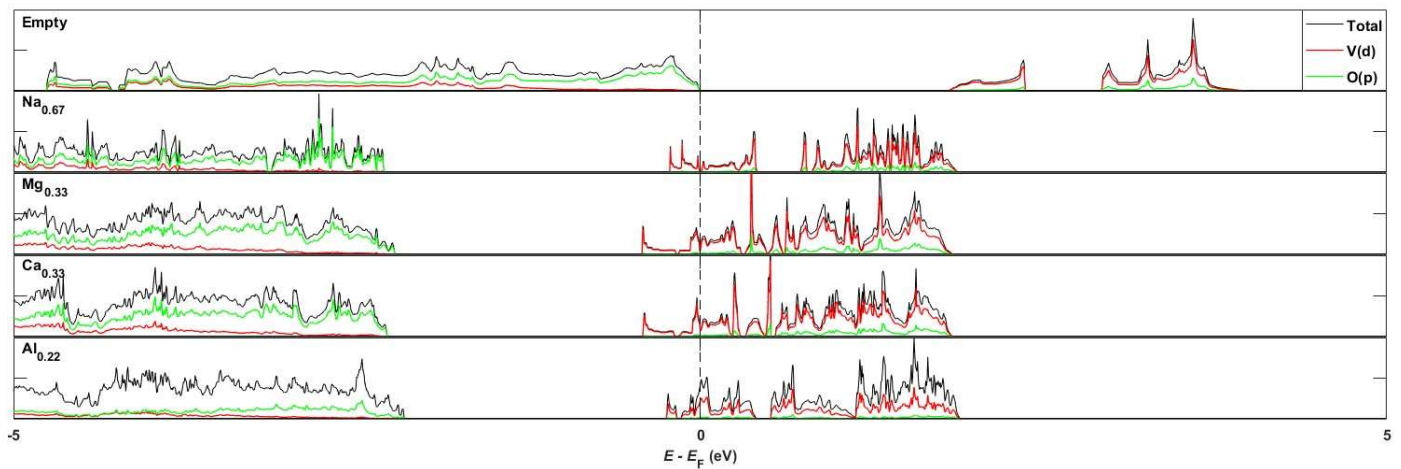


Figure S3: Atom-resolved DOS for Na, Mg, Ca and Al intercalated in α -V₂O₅ using PBE. The total DOS are plotted in black, vanadium *d*-states in red, and oxygen *p*-states in green. The *x*-axis is zoomed to $[-5, 5]$ and adjusted so that “0” corresponds to the Fermi energy, denoted by the dashed line.

S2.3. Voltage

Computational works [5, 6, 8, 22–25] have reported voltages for α -V₂O₅ intercalated with the ions studied in this investigation. These reported voltages are included in Table S4. Those for Li lie in agreement with those calculated in this work, except the Li_{0.75} & Li_{1.5} values of Braithwaite et al. [22]. Their lower voltage is surprising considering their methods match those of this work. The lower concentrations agree with experiment but become further spread as concentration increases, up to 0.8 V at Li_{1.50}. Na voltages are also in agreement with the present work, except the Na_{0.17} & Na_{0.33} concentrations of Wang et al. [24] (> 0.5 V lower). However, these are in excellent agreement with the experimental work of Ali et al. [26]. Interestingly, DFT overestimates the voltages for Na as opposed to underestimating for Li. Every Mg voltage is in excellent agreement with our values and experiment, with the exception of our PBE Mg_{0.33} value. This outlying value is most likely due to the structural deformities that occurred for this particular calculation. The V₂O₅ layers began to bow, resulting in VO₅ polyhedra disconnecting upon further intercalation. Others' voltages are also in agreement for Ca concentrations, excluding the Ca_{0.056} values of Carrasco [5]. Carrasco's Ca voltages are > 0.5 V lower than others, which are above experiment yet in good agreement. Finally, Al voltages are difficult to compare due to the scarcity of available data. However, the Al_{0.05} concentration of Kulish et al. [6] is closest to that used in this investigation ($x = 0.10$) and results in 0.43 V higher than this work.

M_x	Voltage (V)	Type of work	Ref.
Li (OCV)	3.35	DFT (PBE+ U)	[8],[25]
Li _{0.05}	3.13	DFT (PBEsol)	[6]
Li _{0.056}	3.24	DFT (PBE)	[5]
Li _{0.056}	3.50	DFT (optPBE-vdW)	[5]
Li _{0.25}	3.4	Exp.	[27],[28],[29]
Li _{0.25}	3.0	DFT (PBE)	[22]
Li _{0.75}	3.2	Exp.	[27],[28],[29]
Li _{0.75}	1.8	DFT (PBE)	[22]
Li _{1.5}	2.3	Exp.	[28],[29]
Li _{1.50}	1.5	DFT (PBE)	[22]
Li _{1.75}	2.3	Exp.	[27]
Li _{2.0}	2.1	Exp.	[28]
Li _{2.25}	2.0	Exp.	[29]
Na (OCV)	3.49	DFT (PBE+ U)	[8]
Na _{0.05}	3.17	DFT (PBEsol)	[6]
Na _{0.056}	3.32	DFT (PBE)	[5]
Na _{0.056}	3.62	DFT (optPBE-vdW)	[5]
Na _{0.17}	2.85	DFT (PBE+ U)	[24]
Na _{0.25}	2.1	Exp.	[27]
Na _{0.33}	2.70	DFT (PBE+ U)	[24]
Na _{0.5}	2.6	Exp.	[26]
Na _{0.5}	2.0	Exp.	[27]
Na _{0.67}	2.56	DFT (PBE+ U)	[24]
Na _{1.0}	2.0	Exp.	[26]
Na _{1.0}	2.29	DFT (PBE+ U)	[24]
Na _{1.5}	1.4	Exp.	[27]
Mg (OCV)	2.59	DFT (PBE+ U)	[8],[25]
Mg _{0.05}	2.4	Exp.	[30]
Mg _{0.05}	2.75	DFT (PBEsol)	[6]
Mg _{0.056}	2.50	DFT (PBE)	[5]
Mg _{0.056}	2.70	DFT (optPBE-vdW)	[5]
Mg _{0.33}	1.4	Exp.	[31]
Ca (OCV)	3.38	DFT (PBE+ U)	[23]
Ca _{0.056}	2.46	DFT (PBE)	[5]
Ca _{0.056}	2.79	DFT (optPBE-vdW)	[5]
Ca _{0.33}	3.26	DFT (PBE+ U)	[24]
Ca _{0.50}	2.8	Exp.	[32]
Ca _{0.67}	3.02	DFT (PBE+ U)	[24]
Ca _{1.0}	2.25	DFT (PBE+ U)	[24]
Al (OCV)	1.59	DFT (PBE+ U)	[8]
Al _{0.05}	1.70	DFT (PBEsol)	[6]

Table S4: Reported voltages for various M_x concentrations in α -V₂O₅ from experimental and computational works.

The raw data (i.e., take $n - m = 1$ in Eqn. 2) for Na intercalation into $\alpha\text{-V}_2\text{O}_5$ displays a drastic change in the profile at a concentration of $\text{Na}_{1.25}\text{V}_2\text{O}_5$ (See Fig. S4 below). This concentration corresponds to the fifth intercalated ion out of the eight found possible. Due to this, the formation energy for each concentration was determined (using Eqn. S1) and a convex hull (Fig. S5) was created. As suspected, this and the following two concentrations are above the hull and are therefore excluded from the final voltage profile (Fig. 2). Coincidentally, the final structure of these three data points were the ones that deviated the most from the orthorhombic structure (e.g., $\geq 1.25^\circ$ for lattice angles).

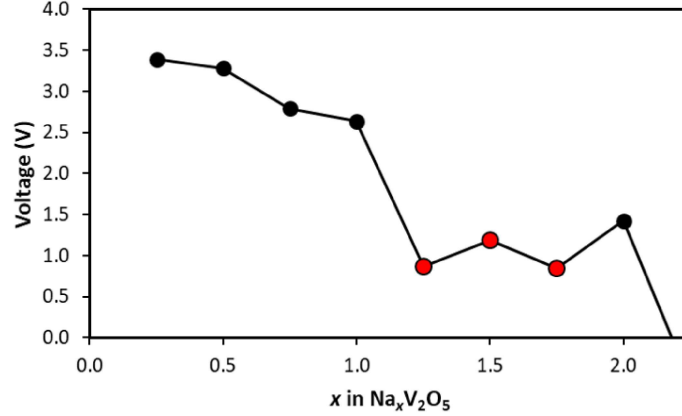


Figure S4: Raw data for Na voltage profile in $\alpha\text{-V}_2\text{O}_5$. NOTE: The concentrations of $x = 1.25, 1.5$ & 1.75 (shown in red) are above the hull (Fig. S5) and excluded from the final voltage profile (Fig. 2).

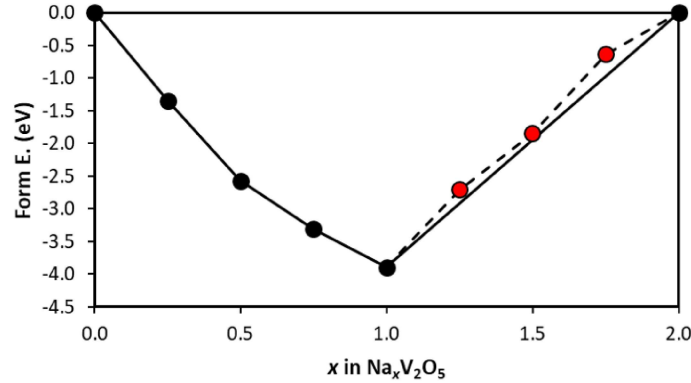


Figure S5: Convex hull for Na voltage profile in $\alpha\text{-V}_2\text{O}_5$. NOTE: The concentrations of $x = 1.25, 1.5$ & 1.75 (shown in red) are above the hull and excluded from the final voltage profile (Fig. 2).

$$E_{\text{Form}} = E_{\text{Na}_x\text{V}_8\text{O}_{20}} - xE_{\text{NaV}_8\text{O}_{20}} - (1 - x)E_{\text{V}_8\text{O}_{20}} \quad (\text{S1})$$

In the above equation, $E_{\text{Na}_x\text{V}_8\text{O}_{20}}$ is the energy of the partially intercalated material, $E_{\text{NaV}_8\text{O}_{20}}$ is the energy of the fully intercalated material, and $E_{\text{V}_8\text{O}_{20}}$ is the energy of the empty structure. It should be noted that the concentration of intercalation, x , is relative to the fully intercalated concentration (i.e., 8 Na ions).

The phonon band structure, Fig. S6, was calculated to further investigate dynamic stability of $\alpha\text{-V}_2\text{O}_5$. The supercell approach in the PHONOPY package [33, 34] was employed with a $2 \times 2 \times 1$ supercell. Stability was held to a < 0.3 THz criterion for imaginary frequency amplitudes to accommodate numerical errors inherent in typical DFT phonon calculations [35–37]. As shown below, all frequencies calculated are within this threshold and demonstrate the stability of $\alpha\text{-V}_2\text{O}_5$.

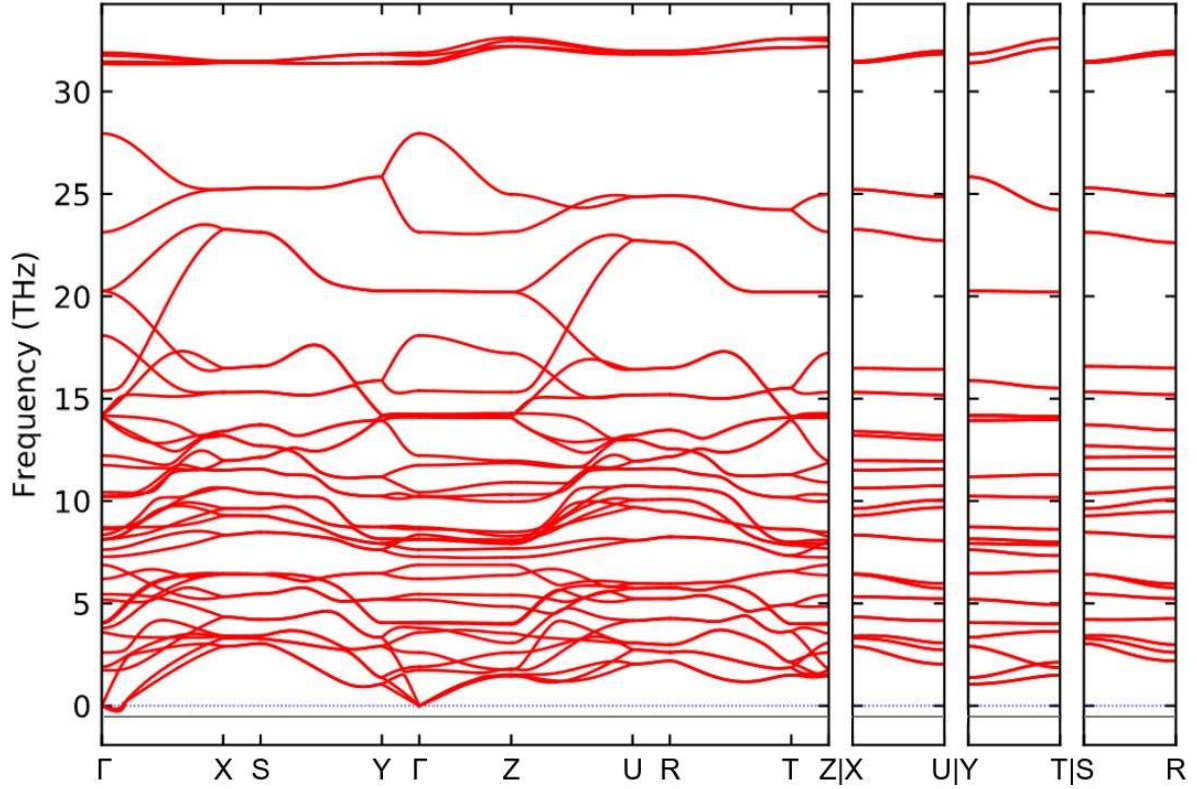


Figure S6: Calculated phonon band structure for the empty $\alpha\text{-V}_2\text{O}_5$ structure. For convenience, phonon bands with imaginary frequencies are shown as negative frequencies. A solid black line represents the numerical error of 0.3 THz within typical DFT phonon calculations [35–37].

For intercalated structures, Fig. S5 & Fig. S6 are used as justification of intermediate stability. However, with the scope of the current work and computational cost, the band structure for fully sodiated $\alpha\text{-V}_2\text{O}_5$, empty and intercalated $\zeta\text{-V}_2\text{O}_5$ was not calculated.

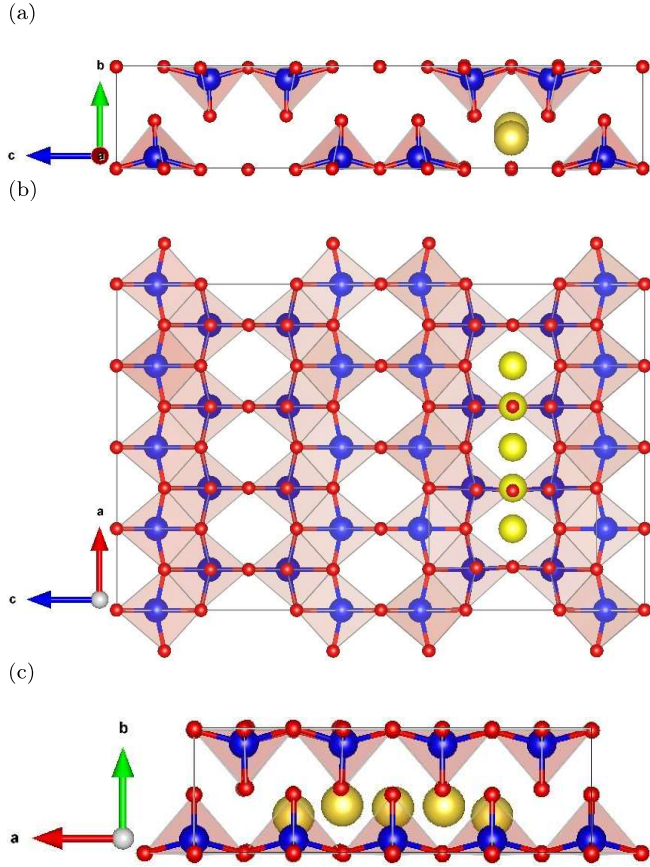


Figure S7: The ion migration pathway along the a vector (lowest energy pathway) for α - V_2O_5 viewed along the (a) a , (b) b , and (c) c lattice vector.

Method	Functional	Barrier (eV)	Ref.
<u>Li</u>			
NEB	PBE	0.16	[8]
NEB	PBE	0.12	[38]
NEB	PBE+ U ($U = 4$)	0.35	[4]
NEB	PBE-D3	0.39	^a
NEB	TS	0.22	[21]
CI-NEB	PBE	0.15	[5]
CI-NEB	PBE+ U ($U = 3$)	0.21	[6]
CI-NEB	PBE@optPBE-vdW	0.24	[5]
CI-NEB	optPBE-vdW	0.31	[5]
<u>Na</u>			
NEB	PBE	0.94	[8]
NEB	PBE-D3	1.27	^a
CI-NEB	PBE	0.85	[5]
CI-NEB	PBE+ U ($U = 3$)	1.06	[6]
CI-NEB	PBE@optPBE-vdW	1.06	[5]
CI-NEB	optPBE-vdW	1.16	[5]

Table S5: Reported ion diffusion barriers for Li and Na in α - V_2O_5 from this work^a and other computational works.

Method	Direction	Functional	Barrier (eV)
<u>Li</u>			
NEB	<i>b</i> vector	PBE-D3	N/A
NEB	<i>c</i> vector	PBE-D3	$\mathcal{O}(10^3)$
<u>Na</u>			
NEB	<i>b</i> vector	PBE-D3	N/A
NEB	<i>c</i> vector	PBE-D3	$\mathcal{O}(10^3)$

Table S6: Ion diffusion barriers for Li and Na along the *b* and *c* lattice vectors (non-lowest energy pathways) in α -V₂O₅. NOTE: The calculations presented here were not completed (i.e., convergence achieved) as barriers were noticeably higher than those shown in Table S5. No barrier is presented for the *b* vector direction as calculations would not run due to small interatomic distances which arose due to the channel in this direction being smaller than the ions used (images are ball-and-stick, not space filling) causing oxygen atoms to overlap with migrated ion species.

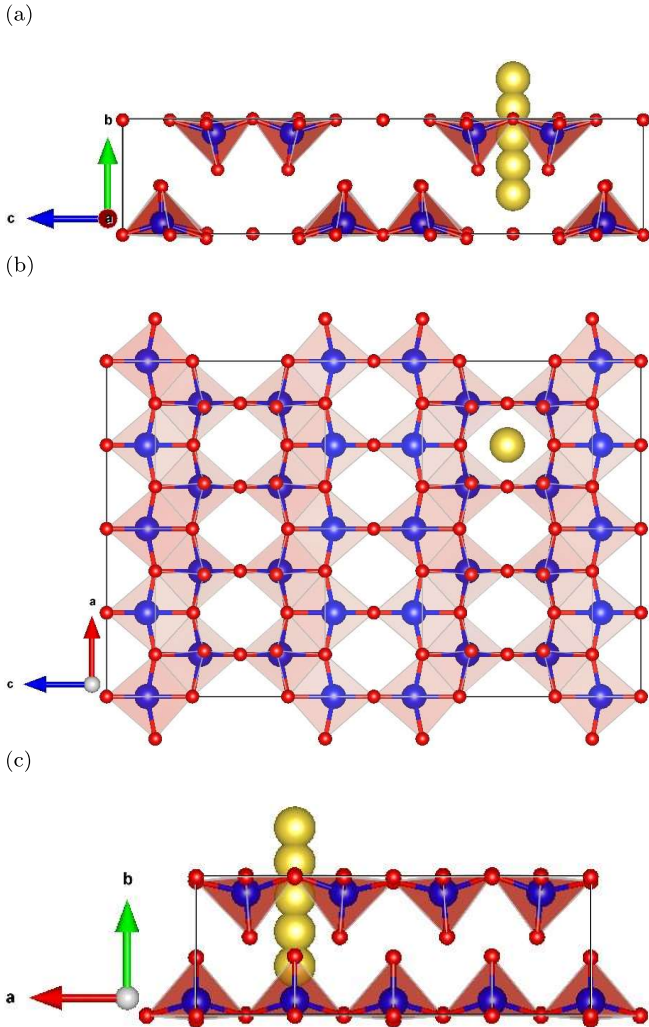


Figure S8: The ion migration pathway along the *b* vector for α -V₂O₅ viewed along the (a) *a*, (b) *b*, and (c) *c* lattice vector.

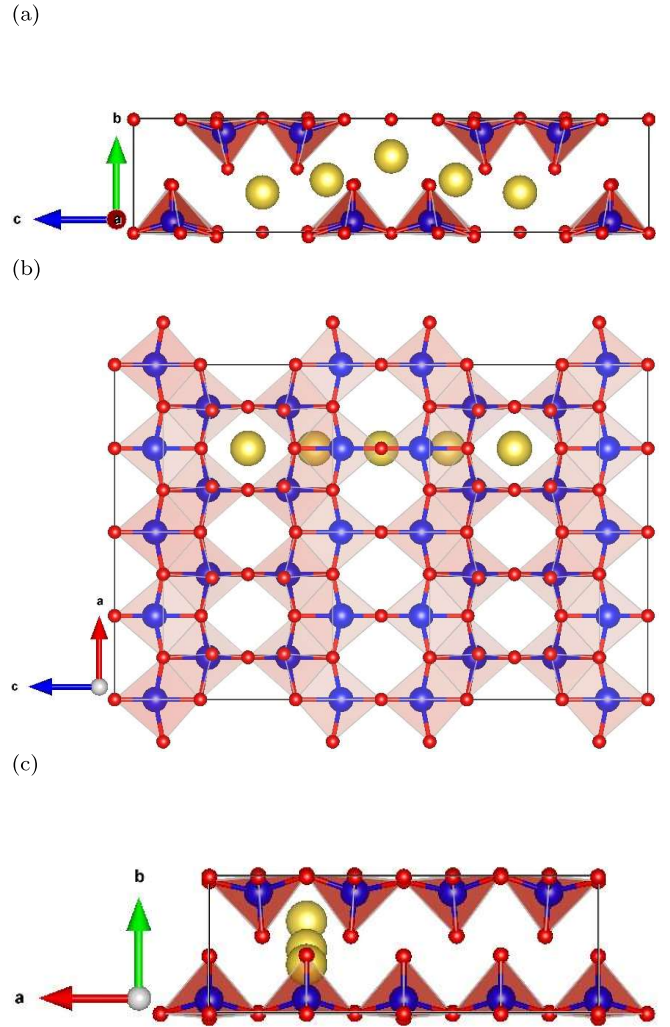


Figure S9: The ion migration pathway along the *c* vector for α -V₂O₅ viewed along the (a) *a*, (b) *b*, and (c) *c* lattice vector.

Section S3:. ζ -V₂O₅ Empty Properties

S3.1. *Physical*

Only one other computational work [8] investigated the empty structure of ζ -V₂O₅ and reported lattice parameters. These results are listed below with those of this work. Unfortunately, there exists no direct match to the methods of this investigation and the other computational work, therefore no intended direct comparison is implied. The ranges for computational data for a , b , and c lattice parameters are 15.406–15.760 Å, 3.585–3.696 Å, and 10.192–10.394 Å, respectively. Comparing these ranges to experimental data in Table 3, one can see a and c are slightly overestimated using DFT and b agrees quite well.

Functional Level	Lattice Parameter			Ref.
	a(Å)	b(Å)	c(Å)	
PBE	15.760	3.585	10.305	^a
PBE+ U (4.0 eV)	15.586	3.666	10.269	[8]
vdW-D2	15.735	3.696	10.394	[8]
PBE-D3	15.406	3.586	10.192	^a

Table S7: Lattice parameters for ζ -V₂O₅ from this work^a and other computational works.

S3.2. Electronic

Partial or atom-resolved DOS were completed for the empty ζ -V₂O₅ structure using the xc functionals and vdW corrections listed in Sec. 1, with the corresponding bandgaps for each plot shown in Fig. 4b. When comparing the PBE and the PBE+ U plots, an expected steady increase in bandgap can be seen. What is unexpected is how the meta-GGA functional of SCAN results in bandgaps in agreement with standard GGA functionals. Based on the behavior of the higher level of theory functional, later electronic calculations (i.e., those for intercalated systems) were investigated with only PBE and PBE-D3.

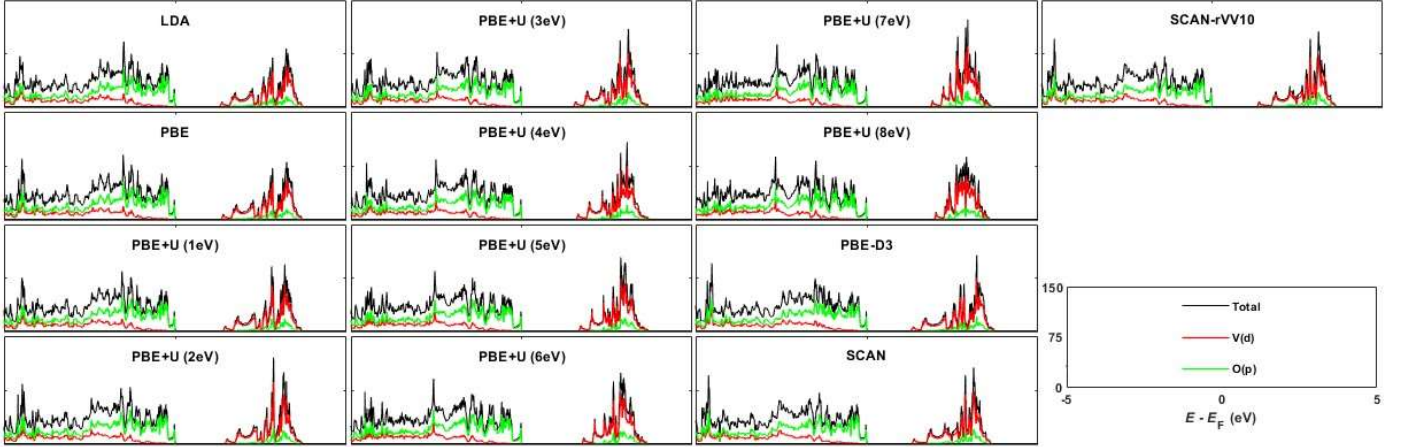


Figure S10: Atom-resolved DOS for empty ζ -V₂O₅ using various functionals. The total DOS are plotted in black, vanadium d -states in red, and oxygen p -states in green. The x -axis is zoomed to $[-5,5]$ and adjusted so that “0” corresponds to the Fermi energy.

Functional	Bandgap (eV)	Ref.
Exp.	1.5	[17]
Exp.	1.1	[39]
LDA	1.745	^a
PBE	1.350	^a
PBE+ U ($U = 1.0$ eV)	1.405	^a
PBE+ U ($U = 2.0$ eV)	1.460	^a
PBE+ U ($U = 3.0$ eV)	1.530	^a
PBE+ U ($U = 4.0$ eV)	1.605	^a
PBE+ U ($U = 5.0$ eV)	1.685	^a
PBE+ U ($U = 6.0$ eV)	1.775	^a
PBE+ U ($U = 7.0$ eV)	1.875	^a
PBE+ U ($U = 8.0$ eV)	1.980	^a
PBE-D3	1.280	^a
SCAN	1.320	^a
SCAN-rVV10	1.320	^a

Table S8: Reported bandgaps for empty ζ -V₂O₅ from this work^a, as well as experimental works.

Section S4:. ζ -V₂O₅ Intercalated Properties

S4.1. Physical

The lattice parameters for mono-, di-, and trivalent intercalation into ζ -V₂O₅ (as described in Sec. 2.2) are listed in Table S9. Li_{0.17} and Al_{0.20} have very similar b lattice parameters after relaxation with both functionals and c with PBE-D3. However, a differs by 0.67 Å and 0.97 Å for the two functionals, respectively, and c by 0.13 Å for PBE. For the $x = 0.33$ concentrations, Li and Ca significantly differ in a (0.58 Å and 0.86 Å) for the two functionals, while Li and Mg differ only in a and c (0.43 Å and 0.14 Å, respectively) for PBE-D3. Li_{0.67} and Na_{0.67} differ in a and c with both functionals. Lattice parameter c differs by 0.13 Å for both functionals and a by 0.10 Å for PBE. When using PBE-D3, a differs by 0.76 Å for these two intercalants.

No clear trend in lattice parameters is evident through Li intercalation concentrations. However, one similarity emerges when comparing each concentration to the empty structure rather than the preceeding concentration. For both PBE and PBE-D3, c is always smaller upon intercalation. Lattice parameter a and b may change similarly at identical concentrations between both functionals, but no distinct trend is present.

In addition to the previously mentioned parameters, corresponding cell volumes are recorded in Table S9. The Li concentrations decrease in volume first, but then follow the trend of higher concentrations having larger volumes for PBE-D3. This trend occurs for PBE as well, but Li_{0.67} decreases from the previous concentration. Comparing intercalations of the same concentration shows that both functionals show larger ions result in larger volumes, with the exception of Mg when using PBE. It is difficult to determine if Al adheres to this trend due to its smaller size than Li but higher concentration.

M _x	Lattice Parameter				
	a(Å)	b(Å)	c(Å)	$\beta(^{\circ})$	Vol.(Å ³)
V ₁₂ O ₃₀ (PBE)	15.760	3.585	10.305	110.400	545.738
Li _{0.17}	15.671	3.589	10.231	110.466	539.073
Li _{0.33}	16.100	3.590	10.168	112.230	544.076
Li _{0.50}	16.354	3.581	10.189	113.637	546.581
Li _{0.67}	15.733	3.595	10.226	110.315	542.359
Na _{0.67}	15.837	3.612	10.353	107.298	565.491
Mg _{0.33}	16.119	3.597	10.131	113.816	537.290
Ca _{0.33}	16.677	3.615	10.127	113.204	561.183
Al _{0.20}	16.343	3.614	10.102	115.106	540.259
V ₁₂ O ₃₀ (PBE-D3)	15.406	3.586	10.192	110.469	527.541
Li _{0.17}	15.373	3.588	10.116	110.429	522.847
Li _{0.33}	15.499	3.581	10.162	111.839	523.582
Li _{0.50}	16.051	3.579	10.096	113.211	533.066
Li _{0.67}	15.573	3.611	10.146	110.645	533.888
Na _{0.67}	15.384	3.615	10.297	107.095	547.413
Mg _{0.33}	15.931	3.594	10.025	113.128	527.827
Ca _{0.33}	16.354	3.623	10.028	112.386	549.331
Al _{0.20}	16.343	3.614	10.102	115.106	518.224

Table S9: Lattice parameters for the various M_x concentrations in ζ -V₂O₅ when using PBE and PBE-D3. All structures remained monoclinic. NOTE: concentrations are relative to V₂O₅, yet lattice parameters are relative to the V₁₂O₃₀ unit cell as shown in Fig. 4a.

S4.2. Electronic

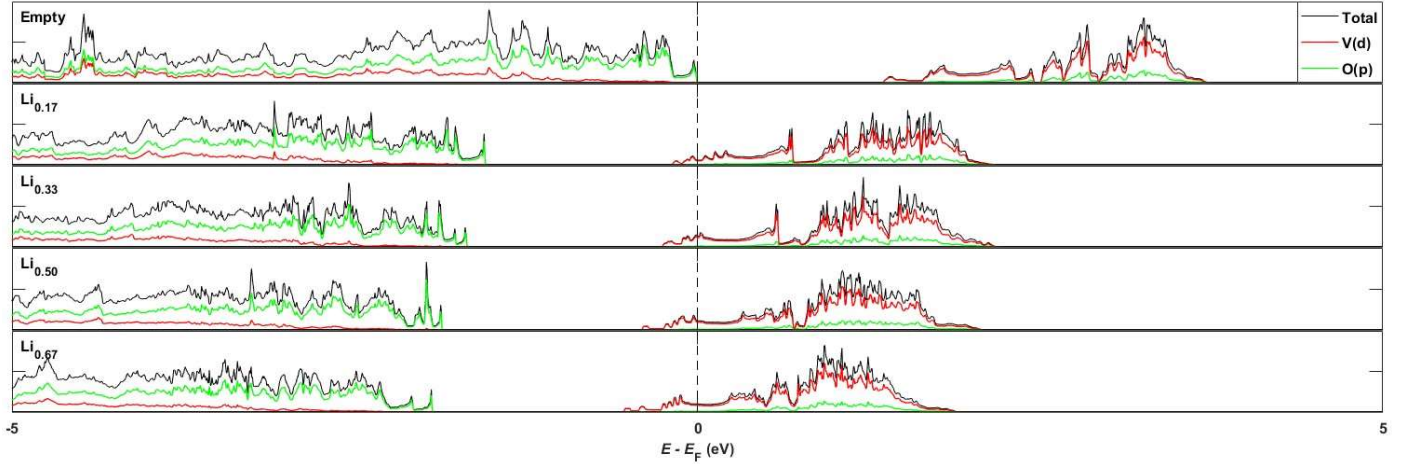


Figure S11: Atom-resolved DOS for Li intercalated in ζ -V₂O₅ using PBE. The total DOS are plotted in black, vanadium *d*-states in red, and oxygen *p*-states in green. The *x*-axis is zoomed to $[-5, 5]$ and adjusted so that “0” corresponds to the Fermi energy, denoted by the dashed line.

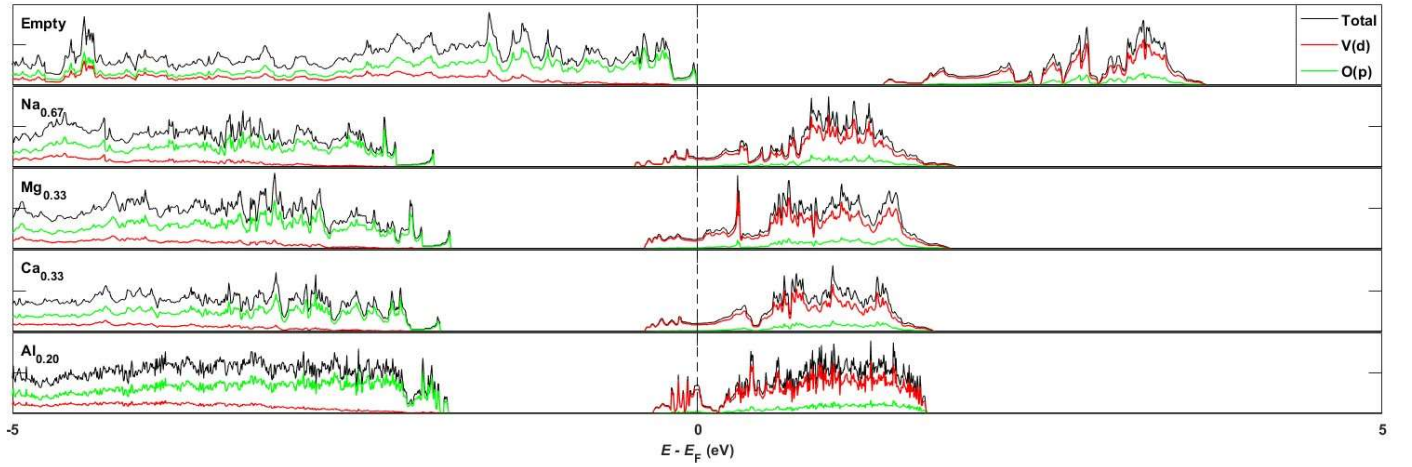


Figure S12: Atom-resolved DOS for Na, Mg, Ca and Al intercalated in ζ -V₂O₅ using PBE. The total DOS are plotted in black, vanadium *d*-states in red, and oxygen *p*-states in green. The *x*-axis is zoomed to $[-5, 5]$ and adjusted so that “0” corresponds to the Fermi energy, denoted by the dashed line.

S4.3. Voltage

One other computational study [8] with a reported voltage for ζ - V_2O_5 , in the context of this work, was found. This work, along with experimental works, is listed in Table S10.

M_x	Voltage (V)	Type of work	Ref.
Li (OCV)	3.7	Exp.	[28]
$\text{Li}_{0.75}$	2.9	Exp.	[28]
$\text{Li}_{1.5}$	2.6	Exp.	[28]
$\text{Li}_{2.0}$	2.2	Exp.	[28]
Na (OCV)	3.94	DFT (PBE+ U)	[8]
$\text{Na}_{0.25}$	3.4	Exp.	[40]
$\text{Na}_{1.0}$	2.9	Exp.	[40]
$\text{Na}_{1.5}$	2.6	Exp.	[40]

Table S10: Reported voltages for various M_x concentrations in ζ - V_2O_5 from experimental and computational works.

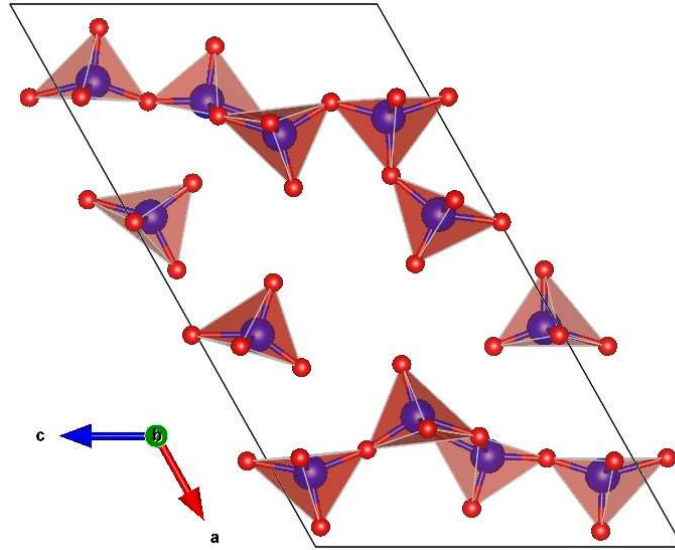


Figure S13: The relaxed structure for the $\text{Na}_{1.33}$ concentration in ζ - V_2O_5 . For ease of viewing the appearing α - V_2O_5 , all Na atoms have been removed. Vanadium atoms are shown in blue and oxygen in red.

The raw data (i.e., take $n - m = 1$ in Eqn. 2) for Na intercalation into ζ - V_2O_5 displays a drastic change in the profile at a concentration of $\text{Na}_{1.00}\text{V}_2\text{O}_5$ (See Fig. S14 below). This concentration corresponds to the sixth intercalated ion out of the eight found possible. Due to this, the formation energy for each concentration was determined (using Eqn. S1) and a convex hull (Fig. S15) was created. As suspected, this and the following concentration are above the hull and are therefore excluded from the final voltage profile (Fig. 5). As it turns out, the final structure of these data points and the $x = 1.33$ were where the V_2O_5 structure began the unzipping mentioned in Sec. 3.9. Due to the extent of this irreversible structural transformation, further intercalation was not completed.

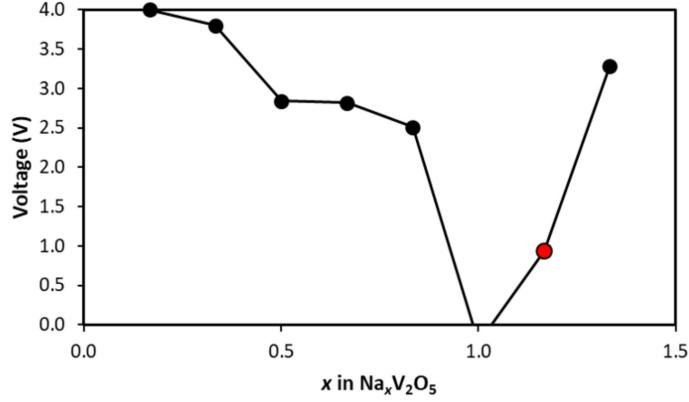


Figure S14: Raw data for Na voltage profile in ζ - V_2O_5 . NOTE: The concentrations of $x = 1.0$ & 1.17 (shown in red) are above the hull (Fig. S15) and excluded from the final voltage profile (Fig. 5).

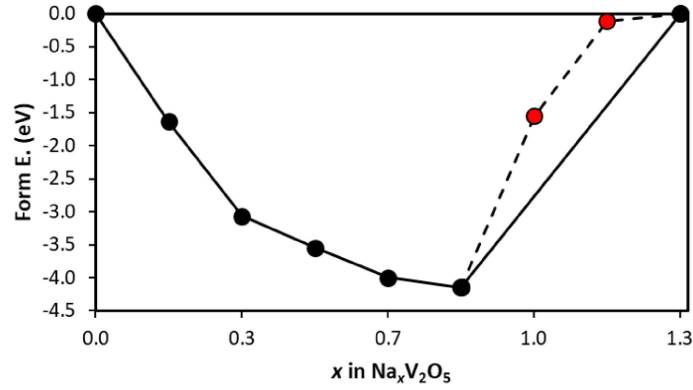
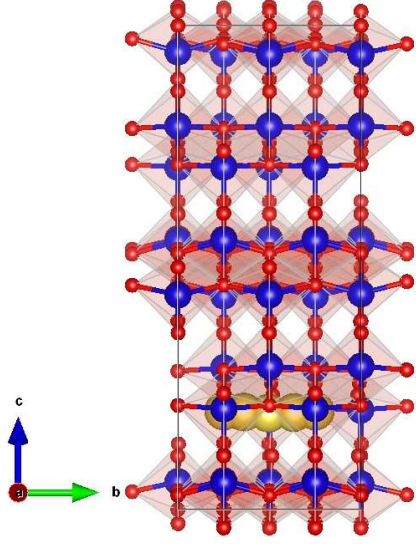


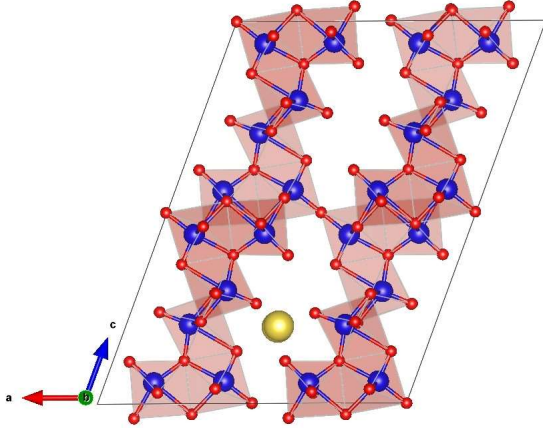
Figure S15: Convex hull for Na voltage profile in ζ - V_2O_5 . NOTE: concentrations of $x = 1.0$ & 1.17 (shown in red) are above the hull and excluded from the final voltage profile (Fig. 5).

S4.4. Nudged Elastic Band - Diffusion Energy Barrier

(a)



(b)



(c)

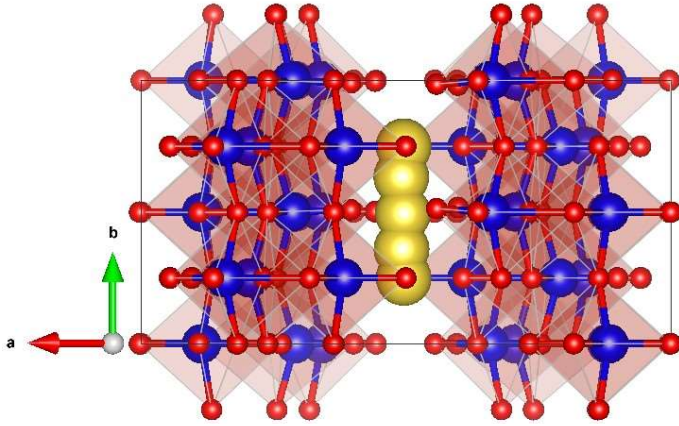


Figure S16: The lowest energy pathway for ζ -V₂O₅ viewed along the (a) *a*, (b) *b*, and (c) *c* lattice vector.

Method	Functional	Barrier (eV)	Ref.
<u>Li</u>			
NEB	PBE	0.11	[8]
NEB	PBE-D3	0.32	^a
<u>Na</u>			
NEB	PBE	0.38	[8]
NEB	PBE-D3	1.09	^a

Table S11: Reported ion diffusion barriers for Li and Na in ζ -V₂O₅ from this work^a and another computational work.

Due to the behavior of migration pathways investigated in α -V₂O₅, ion migration along the *a* and *c* lattice vectors were not investigated for ζ -V₂O₅. Similar behavior is expected due to intercalation channel size and interatomic distances.

Section S5:. Structure Coordinates (POSCAR files)

POSCAR1

```

Empty Alpha V205
1.0
      3.5668363571      0.0000000000      0.0000000000
      0.0000000000      4.7026276588      0.0000000000
      0.0000000000      0.0000000000      11.5497083664
V      0
4      10
Direct
      0.0000000000      0.109795459      0.148374051
      0.5000000000      0.890204549      0.351625979
      0.5000000000      0.890204549      0.648374021
      0.0000000000      0.109795459      0.851625979
      0.5000000000      0.007314556      0.181684300
      0.0000000000      0.992685437      0.318315685
      0.0000000000      0.992685437      0.681684315
      0.5000000000      0.007314556      0.818315685
      0.0000000000      0.451700240      0.856170714
      0.5000000000      0.548299730      0.643829286
      0.5000000000      0.548299730      0.356170684
      0.5000000000      0.999706328      0.500000000
      0.0000000000      0.000293681      0.000000000
      0.0000000000      0.451700240      0.143829301

```

Empty Zeta V205

1.0

15.7600326538	0.0000000000	0.0000000000
0.0000000000	3.5850625038	0.0000000000
-3.5921186508	0.0000000000	9.6589663626

V	0
12	30

Direct

0.116245940	0.000000000	0.111013085
0.883754075	0.000000000	0.888986886
0.616245925	0.500000000	0.111013085
0.383754045	0.500000000	0.888986886
0.337794572	0.000000000	0.111869372
0.662205458	0.000000000	0.888130605
0.837794542	0.500000000	0.111869372
0.162205428	0.500000000	0.888130605
0.288972944	0.000000000	0.411923349
0.711027026	0.000000000	0.588076651
0.788972974	0.500000000	0.411923349
0.211027056	0.500000000	0.588076651
0.000000000	0.000000000	0.000000000
0.500000000	0.500000000	0.000000000
0.111583911	0.000000000	0.265151829
0.888416052	0.000000000	0.734848201
0.611583948	0.500000000	0.265151829
0.388416082	0.500000000	0.734848201
0.135701478	0.500000000	0.071246706
0.864298522	0.500000000	0.928753316
0.635701478	0.000000000	0.071246706
0.364298493	0.000000000	0.928753316
0.266037792	0.000000000	0.222846404
0.733962178	0.000000000	0.777153552
0.766037822	0.500000000	0.222846404
0.233962193	0.500000000	0.777153552
0.438459218	0.000000000	0.222924024
0.561540782	0.000000000	0.777075946
0.938459218	0.500000000	0.222924024
0.061540775	0.500000000	0.777075946
0.316633672	0.500000000	0.056372508
0.683366358	0.500000000	0.943627536
0.816633642	0.000000000	0.056372508
0.183366343	0.000000000	0.943627536
0.396907985	0.000000000	0.475788653
0.603092074	0.000000000	0.524211287
0.896907926	0.500000000	0.475788653
0.103092015	0.500000000	0.524211287
0.255503118	0.500000000	0.426057994
0.744496882	0.500000000	0.573941946
0.755503118	0.000000000	0.426057994
0.244496897	0.000000000	0.573941946

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